Group 09

Diogo Dinis, ist199066
diogo.dinis@tecnico.ulisboa.pt

Afonso Lança, ist1103838
afonso.lanca@tecnico.ulisboa.pt

Luís Calado, ist1103883
luis.maria.calado@tecnico.ulisboa.pt

1. INTRODUCTION

This project is motivated by the TREC-COVID Shared Task where users of the system are expected to be "Researchers, clinicians, and policy makers involved with the response to COVID-19 are constantly searching for reliable information on the virus and its impact." The project encompasses exploration of the dataset, unsupervised organization of the collections, exploration of text processing options, implementation of an IR system with boolean, vector space and probabilistic retrieval models, exploration of ranking and evaluation of the implemented IR system, and the modeling of document interdependencies as a way of aiding the IR system.

2. DATASET EXPLORATION

The original dataset is composed of 192509 documents with 1195 duplicate ids, 812572 distinct terms and an average document length of 153.63 terms (when considering title + abstract). After handling the repeated documents (by concatenating all of their characteristics), our dataset contains 191120 distinct documents. On average, each document of the resulting dataset is composed of 107.31 terms, with the largest document having 16472 terms and the smallest one 1 term. By checking the documents content we were able to access that there are 197892 unique terms, the most frequent one being "patient", appearing a total of 206088 times, and the least frequent one "lowprevalence" appearing only 1 time in total. The 10 top terms in the documents, before and after preprocessing, are presented in **Figure 1** and **Figure 2**.

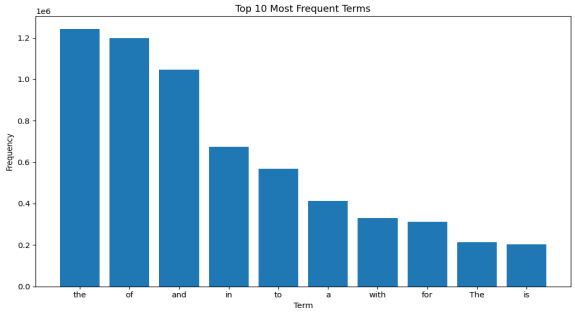


Figure 1. Top 10 terms on corpus before preprocessing

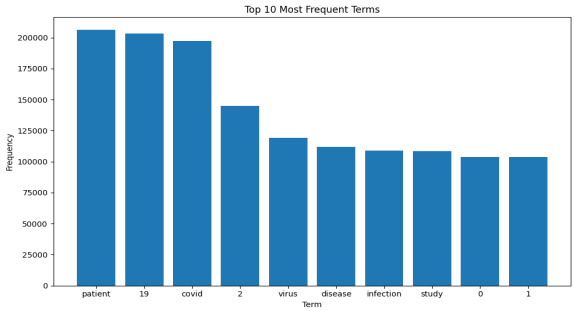


Figure 2. Top 10 terms corpus after preprocessing

Regarding the queries, the original dataset contains 50 queries and 102 unique terms, averaging a length of 3.24 terms, where the longest query contains 7 terms and the smaller one 2 terms. After preprocessing the queries, the quantity of unique terms gets reduced to 88 terms, and, consequently, the biggest query presents now a total of 5 terms, the smaller one 2 terms and, on average the queries now present a length of 2.98 terms. The difference between the top 10 terms before and after preprocessing can be seen on **Figure 3** and **Figure 4**.

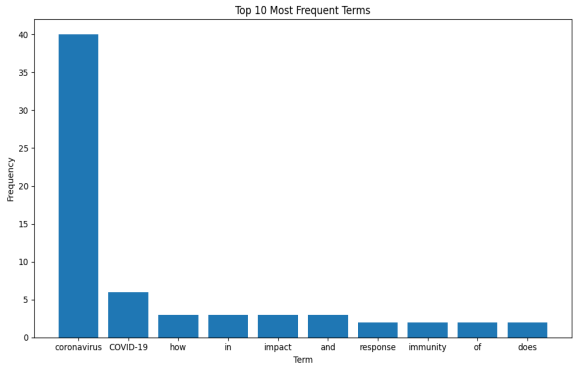


Figure 3. Top 10 terms on queries (Q) before preprocessing

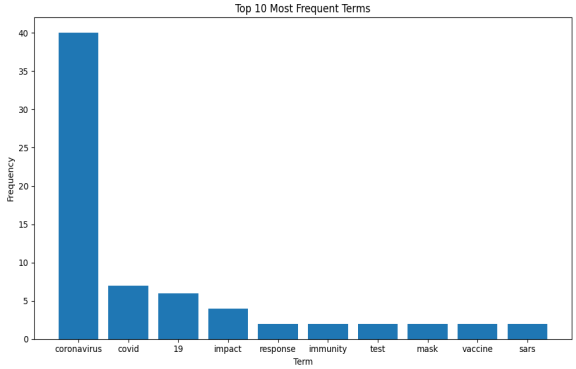


Figure 4. Top 10 terms on queries (Q) after preprocessing

2.1 Unsupervised Collection Organization

Since the dataset provides no a priori categorization for either the document or query collections, unsupervised clustering naturally emerges as the preferred method for estimating their underlying organization. However, most clustering algorithms require specification of the number of clusters (i.e., categories), which here is precisely the parameter to be estimated. The selected approach involves computing clustering solutions over a range of cluster numbers and then comparing these solutions using internal and relative indexes. It is important to note that, due to computational constraints, this analysis was conducted on a subset of documents from the D collection.

2.1.1 *Corpus Organization*

For the corpus collection, TF-IDF was applied to a subset of documents, generating vector space representations that were subsequently clustered using K-Means. The number of clusters was varied between 2 and 30, with the corresponding silhouette coefficients computed and presented in **Figure 5**. In the resulting plot, multiple local maxima are observed as the cluster count increases, each maximum reflecting a different granularity level in the categorization. The chosen range was designed to capture broader category distinctions, ultimately yielding approximately **twenty** categories.

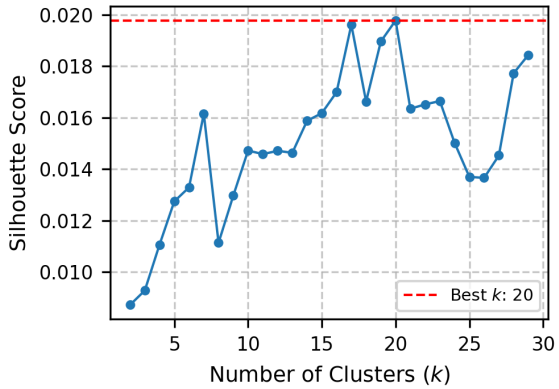


Figure 5. K-Means Clustering of Corpus over k

To evaluate the cohesiveness and separation of the clusters, silhouette coefficients were computed for all documents (see **Figure 5**), being the results presented in **Figure 6**. It is possible to conclude that, although a few clusters exhibit moderate cohesion and separation, most clusters have silhouette scores near zero or even negative, suggesting weak separation.

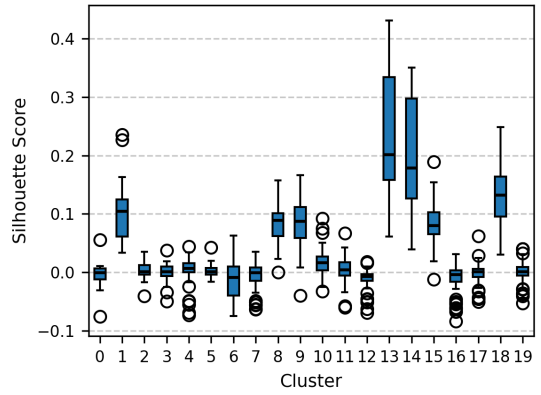


Figure 6. Silhouette Scores per Cluster with k=20

Even so, the clustering reveals that the document collection is organized into themes, ranging from molecular studies and diagnostic assay development to clinical management and public health communication. These thematic distinctions offer insights into how certain queries might interact with the document collection. For instance, the documents relevant for a researcher interested in vaccine development and a policy maker looking for epidemiological data might be encompassed in completely different sets of clusters. Although not explored in this study, this clustering can be used to aid the IR system, for instance by returning the whole cluster if a few of the documents inside it are selected as relevant for a certain topic.

2.1.2 *Queries Organization*

A similar study was conducted for the query collection. However, given the analysis of the query collection performed in the previous section, notably by observation of **Figure 3**, it was decided to first apply lemmatization, followed by removal of terms that appear in more than 5 queries, in an attempt to achieve a better clustering solution.

TF-IDF was then employed to generate vector space representations of the queries, and K-Means clustering was performed with the number of clusters varied between 2 and 50. The corresponding silhouette coefficients, shown in **Figure 7**, suggest that the optimal number of categories is approximately **forty**.

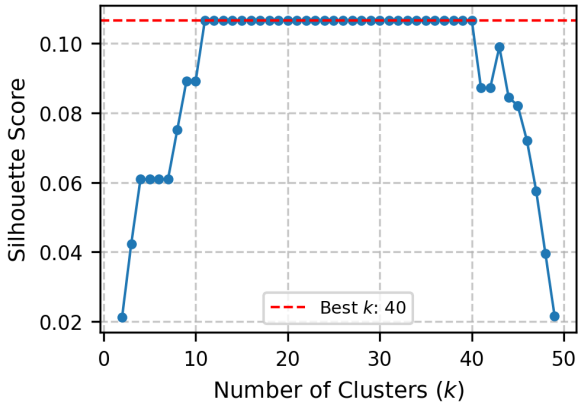


Figure 7. K-Means Clustering of Queries over k

To evaluate cluster cohesion and separation, silhouette coefficients were computed for all queries, being the results presented in **Figure 8**. Initially, only a few clusters appeared moderately cohesive, with many clusters showing a silhouette score of zero. Upon further investigation, it was determined that these zero-score clusters are singleton clusters, which yield a silhouette coefficient of zero by design as to avoid misinterpretation of the clustering quality. This is not a surprising result given that the average query length is only three terms (and some of these terms were removed during preprocessing), resulting in vastly different vector representations that naturally lead to many singleton clusters.

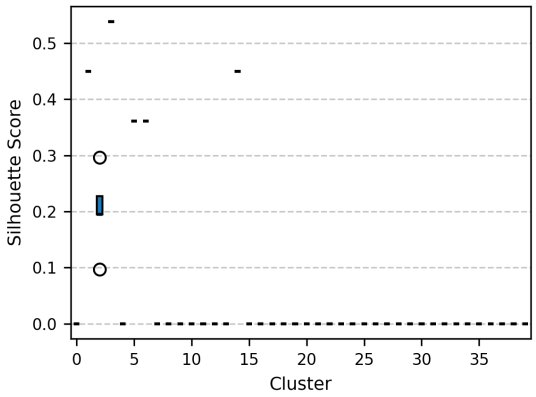


Figure 8. Silhouette Scores per Cluster with k=40

The clustering reveals the query collection to be largely varied conceptually, with many topics represented as individual clusters, although slight overlap in topics is observable in a few of the clusters with non-zero silhouette coefficients. However, further analysis suggests that improvements to the clustering solution are possible. For instance, the queries "Coronavirus outside body" (Cluster 4) and "how long does coronavirus survive on surfaces" (Cluster 30) might be merged, as both pertain to the virus's persistence outside a host. Furthermore, Cluster 5, which groups queries such as "coronavirus social distancing impact", "masks prevent coronavirus", "coronavirus heart impacts", "impact of masks on coronavirus transmission", and "coronavirus mental health impact", includes topics that, while all related to health, differ substantially in focus. This is further underscored by the cluster medoid "impact mask coronavirus transmission", which does not fully encapsulate the varied information needs represented within the cluster.

In an effort to enhance query clustering, query expansion was implemented using the WordNet module from the NLTK library. Unfortunately, the results were not satisfactory, mainly due to the general-purpose WordNet not performing well in this domain. For example, the query "how does coronavirus spread" was incorrectly expanded to "spreading Department of Energy spread doe how gap coronavirus Energy Department". Although not explored, an alternative might involve employing a domain-specific thesaurus or deriving a thesaurus automatically from the corpus based on term co-occurrence statistics.

3. IR SYSTEM PERFORMANCE

All the execution time and memory space results were obtained from the same machine.

3.1 Processing Performance

Five text processing options were implemented:

- Stopword Removal;
- Normalization;
- Tokenization;
- Lemmatization;
- Stemming (Porter);

All the possible combinations of these options were tested, and the resulting metrics (MAP, nDCG, nDCG@10 and P@10) were analyzed and compared. This led to the conclusion that performing removal of stopwords followed by normalization (lowercasing and non-alphanumeric character removal) and, despite its higher computational cost, lemmatization delivered the highest retrieval effectiveness.

This preprocessing step resulted in **90.7276 seconds** of processing time, and the processed documents occupied **180.86 MB**.

Table 1. Time taken for Processing stage

Stage	Time taken (s)	Space used (MB)
Processing	90.7276	180.86

3.2 Indexing Performance

The inverted index is built using a term-document frequency dictionary, where each term is mapped to a dictionary of documents and their respective term frequencies (TF). The indexing process involved iterating through the dataset, preprocessing the text (as described in the processing phase), and populating the index structure.

This resulted in **9.1878 seconds** to process the dataset, and the final index occupied **1378.98 MB**.

Table 2. Time and Space for Indexing Stage

Stage	Time taken (s)	Space used (MB)
Indexing	9.1878	1378.98

3.3 Retrieval Performance

3.3.1 Model Performance

- *Boolean Model*

In our evaluation of the Boolean retrieval model, we chose to use only the query title for matching. Including the query description and narrative would significantly reduce the number of matches since the Boolean model relies on strict set-based operations. Given its binary nature (where documents must contain all query terms to be retrieved), adding more text would make matches rarer, leading to poorer retrieval performance. The metrics obtained for this model, and the corresponding distributions, are presented in **Table 3** and **Figure 9**, respectively.

Table 3. Mean metrics for Boolean over Q with p=1000

Precision	Recall	F-measure with β=1	Time Per Query (s)
0.3502	0.2205	0.2144	0.0061

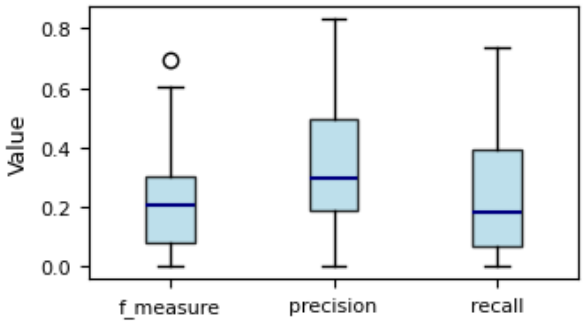


Figure 9. Metric Distributions for Boolean over Q

- *TF-IDF*

The TF-IDF model exhibits many variants, and different weightings are allowed for queries and documents. A few weighting schemes were tested, and the one that yielded the best results metric-wise was **lnn.ntn**. The metrics obtained for this model, and the corresponding distributions, are presented in **Table 4** and **Figure 10**, respectively.

Table 4. Mean metrics for TF-IDF over Q with *p*=1000

MAP	nDCG	nDCG@10	P@10	Time Per Query (s)
0.1878	0.3878	0.5429	0.6380	0.1208

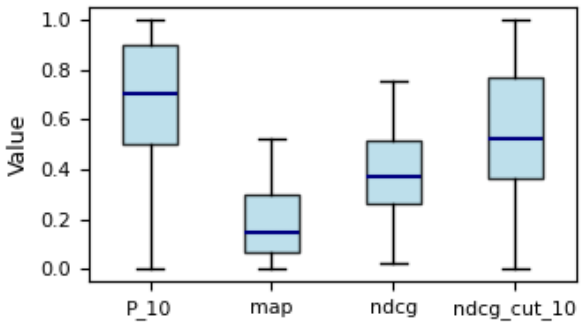


Figure 10. Metric Distributions for TF-IDF over Q with p=1000

- *BM25*

The BM25 model exhibits two parameters, k1 and b, which control the term frequency scaling and the document length normalization, respectively. By trying all the combinations for the values, with k1 between 1 and 2 with a step of 0.1 and b between 0 and 1 with a step of 0.1, it was possible to conclude that **k1=1.1** and **b=0.4** yielded the best results metric-wise. The metrics obtained for this model, and the corresponding distributions, are presented in **Table 5** and **Figure 11**, respectively.

Table 5. Mean metrics for BM25 over Q with *p*=1000

MAP	nDCG	nDCG@10	P@10	Time Per Query (s)
0.2148	0.4153	0.6456	0.7320	0.1259

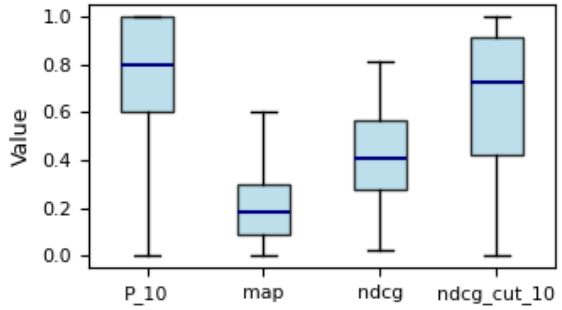


Figure 11. Metric Distributions for BM25 over Q with p=1000

- *Reciprocal Rank Fusion (TF-IDF & BM25)*

Reciprocal Rank Fusion (RRF) can be used for placing consensus between the ranking outputs of TF-IDF and BM25. By leveraging the strengths of each model, RRF helps produce a more balanced ranking that is closer to an optimal retrieval performance. Given a set of queries, we retrieved ranked document lists from each scoring function and fused them using RRF. The metrics obtained for this model, and the corresponding distributions, are presented in **Table 6** and **Figure 12**, respectively.

Table 6. Mean metrics for RRF over Q with *p*=1000

MAP	nDCG	nDCG@10	P@10	Time Per Query (s)
0.2151	0.4299	0.6269	0.7140	0.2453

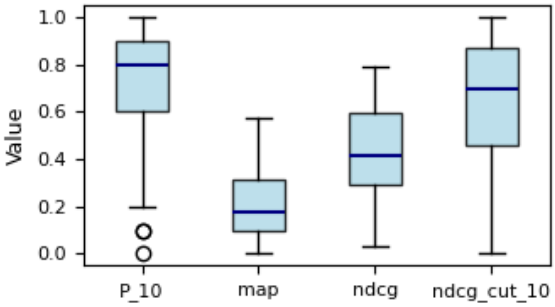


Figure 12. Metric Distributions for RRF over Q with p=1000

3.3.2 Model Comparison

While the precision and recall obtained for the unranked retrieval model, Boolean, can be perceived as unsatisfactory, when comparing it with one of the ranked retrieval models, like BM25, it is possible to conclude that its performance is actually decent, as shown in **Table 7**, while being much faster at retrieval. Its biggest drawback is the lack of ranking, which makes any of the other models more desirable. Experiments were conducted to evaluate the performance of first applying a Boolean retrieval followed by a ranked retrieval on the previously retrieved documents as a way to improve the response time of ranked models, but the low recall exhibited by the Boolean model hindered the overall retrieval performance.

Table 7. Comparison between Boolean and BM25 over Q

	p	Precision	Recall
Boolean	[redacted]	0.3502	0.2205
BM25	1000	0.2094	0.3970

Regarding the ranked retrieval models, which are compared in **Table 8**, it is evident that both BM25 and RRF lead to a better performing IR system. Between BM25 and RRF, we can observe that the RRF model slightly improved MAP and nDCG over BM25, indicating better overall ranking quality. However, nDCG@10 and P@10 are slightly lower, suggesting a reduced effectiveness in top-ranked retrieval. The boxplots further reveal that while RRF achieves a higher median, its lower quartile is significantly worse than BM25, suggesting that some queries perform poorly under this approach. In terms of efficiency, though, this was clearly the lengthiest approach, which is not surprising given that it works as a combination of previous models.

Table 8. Comparison between Ranked Retrieval Models

Model	MAP	nDCG	nDCG@10	P@10	Time Per Query (s)
TF-IDF	0.1878	0.3878	0.5429	0.6380	0.1208
BM25	0.2148	0.4153	0.6456	0.7320	0.1259
RRF	0.2151	0.4299	0.6269	0.7140	0.2453

3.3.3 Best Performing Model - BM25

Depending on the user's preference or use case, more importance is given to either precision or recall. The developed IR system was tested for a range of *p* (number of retrieved documents), and both precision and recall metrics were computed, which are presented in **Table 9**. As expected, if the objective of the user is precision, and thus minimizing false positives, then a lower *p*, under 100, is optimal. If the user's intention is recall, and thus minimizing false negatives and maximizing true positives, then a higher *p* is recommended, preferably equal or higher than 5000. However, it is also possible to observe that, as *p* increases, increase in recall progressively slows down.

Table 9. Precision and Recall with varying p

p	Precision	Recall	FP	FN	TP
10	0.7320	0.0166	2.7	526.0	7.3
100	0.5248	0.1121	47.5	480.8	52.5
500	0.3084	0.3029	345.8	379.1	154.2
1000	0.2094	0.3970	790.6	323.8	209.4
5000	0.0639	0.5791	4680.7	214.0	319.3

From the obtained metrics for this model, **Table 5**, and more notably the relatively low MAP and the relatively high nDCG@10 values obtained, it is possible to conclude that the developed IR System is more adequate for precision purposes. As for the distribution of each metric, **Figure 11**, they exhibit high variance, meaning that the performance of retrieval/ranking does vary considerably between queries.

3.3.4 Aiding IR System with PageRank

The PageRank algorithm is based on representing documents as a graph, where nodes correspond to documents and edges encode relationships between them. This approach utilizes graph centrality metrics to enhance retrieval tasks by modeling the interconnections among documents. It is important to note that in this section a smaller collection version, "cord19/trec-covid/round1", was used.

Given the size of the dataset, and as suggested by the faculty, a PageRank-based reranking system was implemented in an attempt of aiding the IR system comprising the following steps:

1. First stage retrieval: using the BM25 model to retrieve the top 1000 documents for each query.
2. Graph Construction:
    a. Nodes: documents from the top 1000 results.
    b. Edges: represents document similarity, calculated using cosine similarity between the vectors representations of the documents obtained using TF-IDF.
3. PageRank calculation: applies the PageRank algorithm to the graph constructed from the documents.
4. Reranking: the original 1000 documents are re-ranked based on their PageRank scores.

To enhance the effectiveness of the IR system, three different PageRank variants were explored:

- Undirected PageRank (PR)
- Weighted PageRank (WPR): using BM25 scores as priors.
- Mixture model: $\alpha \times WPR + (1 - \alpha) \times BM25$, where α is a parameter that balances the influence of the Weighted PageRank and BM25 scores.

3.3.4.1 PageRank Performance

In terms of performance, the basic PageRank model (PR) performed worse than the BM25 model, indicating that using graph centrality alone—without weighting or priors—is not sufficient for effective document ranking. Performance improved when incorporating BM25-based priors in PageRank (WPR), and further improved when combining weighted PageRank with BM25 scores (WPR + BM25). The metrics obtained for these PageRank variants, and the corresponding distributions, are presented in **Figures 13**, **14**, **15**, and comparison between the different models in **Table 10**. To save computational resources, all graphs were computed a priori so that they could be used for different PageRank implementations. It is important to note that the Time Per Query metric does not account for graph construction time.

Table 10. Comparing BM25 and PageRank variants, θ= 0.1

Model	MAP	nDCG	nDCG @10	P@10	Time Per Query (s)
BM25	0.1901	0.4358	0.4301	0.4933	0.0201
PR	0.0630	0.3043	0.1211	0.1500	5.0202
WPR	0.0790	0.3300	0.1653	0.2067	2.3497
WPR + BM25 α = 0.5	0.1902	0.4360	0.4306	0.4933	2.3850

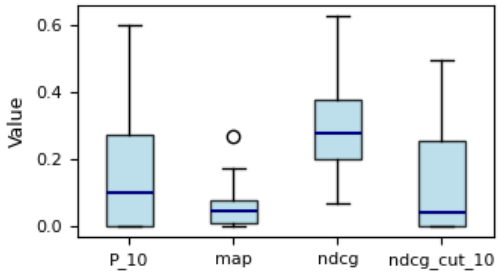


Figure 13. Metric Distributions for PR over Q with p=1000

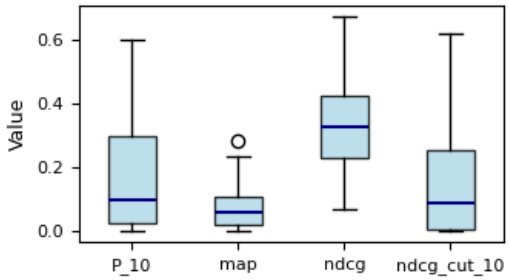


Figure 14. Metric Distributions for WPR over Q with p=1000

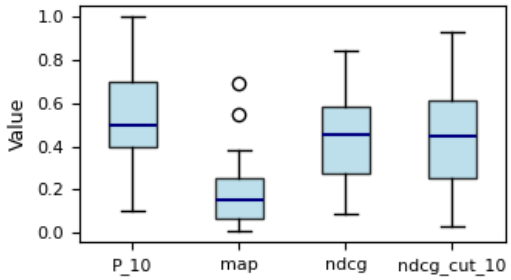


Figure 15. Metric Distributions for WPR + BM25 over Q with p=1000

The use of non-uniform priors based on BM25 scores contributed to the model improvement, as BM25 provides a strong initial relevance estimate that helps guide the PageRank algorithm more effectively. By combining BM25 scores with the structural information from the similarity graph, the ranking process emphasizes documents that are both relevant and central, leading to better results. However, even with these enhancements, the graph-based methods only achieved performance comparable to the BM25 model, while requiring more time per query.

This limited improvement suggests that the graph did not capture strong or meaningful connections between documents. If the graph is too sparse or the documents are only loosely related, PageRank cannot effectively propagate relevance signals. As a result, the graph structure may introduce noise rather than enhance ranking quality, limiting the benefits of combining it with BM25.

3.3.4.2 Calibrating θ for better results

The threshold θ is essential in graph construction as it sets the minimum similarity score needed for an edge between two document nodes. A higher θ leads to a sparser graph, connecting only highly similar documents, which emphasizes relevant relationships but may miss valuable connections. In contrast, a lower θ results in a denser graph, including more edges but potentially introducing noise from less relevant documents, making it challenging to identify central documents.

In an attempt to find the optimal θ, various values of θ were tested to analyze how ranking performance varies with different threshold settings. Concluding that the optimal value of θ is 0.1, given that when θ increases the performance gets worse as visible in **Table 11, 12 and Figure 16**. For the mixture model, different values of alpha were tested; however, the results showed minimal variation, leading to choosing α=0.5. Due to the lengthy run times of the models, testing was conducted with different values of θ across 5 queries instead of the full 30, as using all 30 queries would require computing the graphs 30 × 5 times — once for each combination of query and θ value.

Table 11. Undirected PageRank with different θ values

θ	MAP	nDCG	nDCG@10	P@10
0.1	0.0400	0.1985	0.0990	0.1400
0.3	0.0294	0.1884	0.1136	0.1800
0.5	0.0140	0.1617	0.0405	0.0400
0.7	0.0107	0.1539	0.0000	0.0000
0.9	0.0101	0.1515	0.0000	0.0000

Table 12. Weighted PageRank with different θ values

θ	MAP	nDCG	nDCG@10	P@10
0.1	0.0486	0.2132	0.1503	0.2000
0.3	0.0355	0.1992	0.1197	0.1800
0.5	0.0258	0.1823	0.0410	0.0400
0.7	0.0334	0.1910	0.0327	0.0400
0.9	0.0349	0.1899	0.0000	0.0000

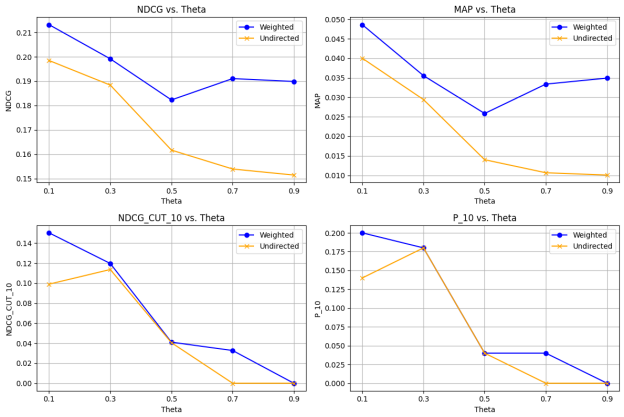


Figure 16. Metric Distributions for PR and WPR varying with θ

3.3.4.3 Documents with Higher Centrality Score

Upon analyzing the graph-based representations of the retrieved documents, it was possible to observe that certain documents consistently exhibited higher centrality scores across multiple queries. These documents ranked highly in measures such as degree and eigenvector centrality, indicating they are well-connected locally or influential globally within the similarity network. For instance, the document with ID *mm4ngrla* attained the highest average degree centrality around 0.42, suggesting frequent connections with other documents and a strong local influence. Likewise, *ky33ju30* achieved the highest average eigenvector centrality around 0.13, implying strong global influence by being connected to other central documents. These patterns are illustrated in **Figures 17 and 18**, which show the average centrality scores across queries for degree and eigenvector centrality, respectively.

By computing average centrality scores across all 30 queries, it was possible to identify a subset of documents that act as central hubs within the document graphs. These central documents may play an important role in bridging topics or encapsulating generally relevant content, making them promising candidates in graph-based ranking strategies. However, further inspection revealed that even the most central document,—*ky33ju30*, according to eigenvector centrality—only appeared as relevant in a limited number of queries (5 out of 30), suggesting that structural centrality alone does not guarantee broad or query-specific relevance. Also the fact that the documents with highest centrality scores still exhibited relatively low scores, reflected in the performance of the PageRank-based reranking system (PR), where the unweighted model underperformed compared to BM25. Improvements were observed only when incorporating BM25-based priors and the combination with BM25 scores, confirming that combining graph structure with relevance signals yields better retrieval results, though the performance was only capable of achieving results similar to using BM25 alone.

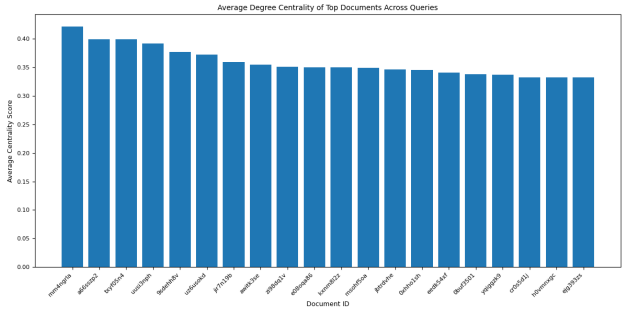


Figure 17. Average Degree Centrality of Top Documents Across Queries

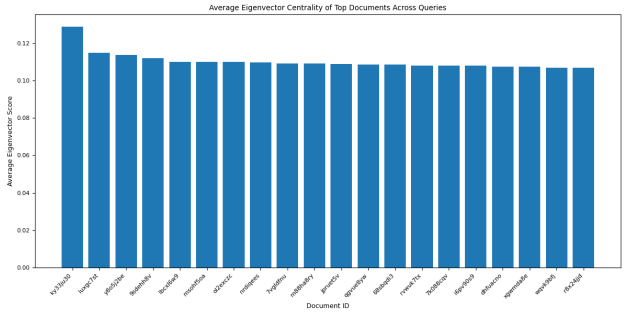


Figure 18. Average Eigenvector Centrality of Top Documents Across Queries

4. CONCLUSIONS

While all the objectives were met, due to time constraints, the team was unable to fully explore certain design choices of the IR system, such as exploring in more depth the preprocessing phase, implementing LM retrieval and integrating the full document text into the IR system as opposed to just using the title and abstract sections. Moreover, given the domain specific nature of the corpus, the target users, and the obtained results, we believe that the current IR System is not the most adequate and could be greatly improved. For the end users, a system with higher MAP and nDCG metrics is likely to be preferable, since they would benefit from the novelty factor of the retrieved documents.